Hints: Abstract algebra – 6. The hardest problems

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. G_p is an increasing union of the subgroups μ_{p^n}, hence a subgroup. A subgroup $K \leqslant G_p$ either contains elements of arbitrarily large p-power order (then $K \supseteq \mu_{p^n}$ for all n, so $K = G_p$) or lies in some μ_{p^N} (then it is finite).

Hint 2. Conversely, let H be infinite with all proper subgroups finite. An element of infinite order would generate an infinite cyclic subgroup, which has the infinite proper subgroup $\langle x^2 \rangle$. So H consists of roots of unity.

Hint 3. Split H into its q-parts H_q (q prime). If infinitely many H_q are non-trivial, dropping one gives an infinite proper subgroup. So only finitely many are non-trivial, and some H_p is infinite, so $H_p = H$. An infinite p-subgroup of the roots of unity contains μ_{p^n} for every n, so $H = G_p$.

Problem 2

Note. "Ring" means ring with unit, as everywhere on these sheets; with non-unital rings one would get $i(\mathbb{Z}_{12}) = 12$.

(a)

Hint 1. A biadditive multiplication on $\mathbb{Z}_n$ is determined by $a = 1 \cdot 1$: $x \cdot y = axy$. It is automatically associative and commutative. It has a unit iff a is invertible mod n. So $i(\mathbb{Z}_n) = \varphi(n)$, and $\varphi(12) = 4$.

(b)

Hint 1. Product rings: a pair of ring structures on A and B gives one on $A \times B$, and different pairs give different multiplications.

(c)

Hint 1. $H_k = \mathbb{Z}_{n_k}$ with $n_k = p_1 p_2 \cdots p_k$: $n_k/\varphi(n_k) = \prod_{i \leqslant k} \frac{p_i}{p_i - 1} \to \infty$, since $\sum 1/p$ diverges.

Hint 2. $G_k = (\mathbb{Z}_2)^k$: transport the ring structure of $\mathbb{F}_2^k$ (or of $\mathbb{F}_{2^k}$) along all group automorphisms $\varphi \in \mathrm{GL}_k(\mathbb{F}_2)$. Two of them give the same multiplication only if they differ by a ring automorphism, and there are at most $k!$ (resp. k) of those. So $i(G_k) \geqslant |\mathrm{GL}_k(\mathbb{F}_2)|/k!$, which is about $2^{k^2}/k!$, far larger than 2^k.

Problem 3

Hint 1. Put $c = f(1)$ and suppose c is even. Take $h = e$: you get $f(n) = f\big(n/\gcd(n, c)\big)$ for every n. In particular $f(2) = f(1) = c$. (If $c = 0$, f is already constant.)

Hint 2. Use a dihedral group: g a rotation of order a, h a reflection. Then $g^2 h^{f(a)}$ is a reflection if $f(a)$ is odd and equals g^2 if $f(a)$ is even, while $h^a g^{f(2)} = h^a g^c$.

Hint 3. For even a: if $f(a)$ were odd you would get $c = f(2) = f(a/\gcd(a, c)) = f(a)$, contradicting parity. So $f(a)$ is even, and then $f(a/2) = f(a)$. Hence $f(a) = f(2a)$ is even for every a.

Hint 4. For odd a (with $f(a)$ even) the same group gives $f(a) = f(2) = c$. So f is constant on odd numbers, and by halving on all numbers, a contradiction.

Problem 4

Hint 1. The sum of all elements of a finite abelian group of exponent p is 0 unless the group is $\mathbb{Z}/2$. So with at least 4 elements the non-zero elements add up to 0. Then for B you need $\sum_{x\in A} x = -\sum_{y\in B} y$.

Hint 2. Try $|B| = 2$, $B = \{b, c\}$: you need $bc = -(b + c)$, i.e. $(1 + b)(1 + c) = 1$.

Hint 3. Take $t \notin \{0, 1, -1\}$ (possible since there are at least 4 elements), $b = t - 1$, $c = t^{-1} - 1$. Then b, c are non-zero and distinct, they commute, and $A = K^* \setminus B$ is non-empty.

Problem 5

(a)

Hint 1. Linearise $[x, f(x)] = 0$ by putting $x + y$.

Hint 2. For the second identity, compute $[x, f(xy)] = [x, f(x)f(y)]$ in two ways: directly, and via the first identity as $[f(x), xy]$. You get $(f(x) - x)[f(x), y] = 0$. Rewrite $[f(x), y] = [x, f(y)]$ and use surjectivity of f.

(b)

Hint 1. In a division ring, (a) says: for each x, either $f(x) = x$ or x commutes with everything. So R is the union of two additive subgroups, $\{x : f(x) = x\}$ and $Z(R)$.

Hint 2. A group is never the union of two proper subgroups. Since $f \neq \mathrm{id}$, $Z(R) = R$.

Problem 6

Hint 1. The identity reads $x(1 - x)^2(1 + x) = 0$. Put $x = 2$: $6 = 0$, and 2 is invertible, so $3 = 0$.

Hint 2. If $n^2 = 0$, the identity gives $n = 0$. So A is reduced. In the commutative subring generated by x, $(x^3 - x)^2$ is a multiple of $x(x - 1)^2(x + 1)$, so $x^3 - x$ is nilpotent, hence 0. Thus $x^3 = x$ for all x.

Hint 3. Then x^2 is idempotent, and idempotents of a reduced ring are central. So $(x + 1)^2 - x^2 - 1 = 2x$ is central, and so is x: A is commutative. Units satisfy $u^2 = 1$.

Hint 4. From $(a + b)^2 = 1 = a^2$ get $b(2a + b) = 0$. If b were a unit, then $b = -2a = a$. And $(1 + ab)^2 = 1 + b(2a + b) = 1$. Example: $\mathbb{F}_3 \times \mathbb{F}_3$.

Problem 7

Hint 1. Sum all the products: in an abelian group $\prod_i (a_i a_{i+1}) = (\prod a_i)^2$ must equal $\prod a_i$. So the sum of all elements must vanish. For $\mathbb{Z}_n$ that sum is $n(n - 1)/2$, which is $\neq 0 \bmod n$ for even n.

Hint 2. For odd n take $a_i = i$ $(i = 0, \ldots, n - 1)$: the sums $2i + 1$ run over all residues, because 2 is invertible.

Hint 3. For even order, look at a group with more than one involution, e.g. $\mathbb{Z}_2 \times \mathbb{Z}_4$. The order $(0, 0), (1, 0), (0, 1), (1, 2), (1, 1), (1, 3), (0, 3), (0, 2)$ works.

Problem 8

Hint 1. Polynomials modulo $x^p - x$ correspond to functions $\mathbb{F}_p \to \mathbb{F}_p$, and composition corresponds to composition. Identify the two generators as functions.

Hint 2. $x+1$ is the cycle $k \mapsto k+1$. For the other polynomial q: if $x \neq 0, 1$ then $q(x) = \frac{x^{p-1}-1}{x-1} + x = x$; also $q(0) = 1$ and $q(1) = 0$. So q is the transposition $(0\,1)$.

Hint 3. A p-cycle and an adjacent transposition generate S_p. In a finite group the generated semigroup is the generated group. So $|W| = p!$.

Problem 9

Hint 1. n is interesting iff $x^n = 1$ in $R = \mathbb{F}_p[x]/(x^p - x + 1)$. So the smallest interesting number is the multiplicative order of x in R.

Hint 2. In R, $x^p = x - 1$, and by the Frobenius $x^{p^k} = x - k$. So $x^{p^p} = x$, and x is invertible, hence $x^{p^p - 1} = 1$.

Hint 3. Norm trick: $x^{1+p+\cdots+p^{p-1}} = \prod_{k \in \mathbb{F}_p}(x-k) = x^p - x = -1$. So $x^{2(p^p-1)/(p-1)} = 1$, which is a smaller interesting number when $p > 3$.

Hint 4. For $p = 2$: x has order 3. For $p = 3$: the order divides 26, $x^2 \neq 1$ and $x^{13} = -1 \neq 1$, so it is 26. Answer: $p \in \{2, 3\}$.

Problem 10

(a)

Hint 1. First show $xa, ax \in D$ for $a \in D$: if ax were a unit, a would have a right inverse, and then $a^2 = 0$ forces $a = 0$.

Hint 2. Show D is closed under addition. If $a + b = u$ is a unit with $a, b \in D$, then $c = u^{-1}a \in D$ and $1 - c = u^{-1}b \in D$, so $0 = (1-c)^2 = 1 - 2c$ and c would be a unit.

Hint 3. Now $a + xa \in D$, so $0 = (a + xa)^2 = a^2 + axa + xa^2 + (xa)^2 = axa$.

(b)

Hint 1. Start with any non-zero $a \in D$. If $ab \neq 0$ for some $b \in D$, replace a by $a' = ab$. Then $a'D \subsetneq aD$: $ab \in aD$, but $ab \notin abD$, since $ab = abd$ would give $ab = abd^2 = 0$. Also $|Da'| = |Dab| \leqslant |Da|$.

Hint 2. Symmetrically for $ba \neq 0$. So $|aD| + |Da|$ strictly decreases while a stays non-zero. By finiteness the process stops at a non-zero a with $aD = Da = 0$.